

Evaluating the Impact of a Cardiovascular Disease Prevention Program on Community Health

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INTRODUCTION

Cardiovascular disease (CVD) is a major cause of death in Sri Lanka and globally which happens due to disorder of heart and blood vessels. Coronary heart disease, heart failures, heart attacks, strokes and irregular heart rhythms are some examples for cardiovascular diseases. According to WHO in 2022 ,32% of global deaths happened due to cardiovascular disease among those deaths 85% were happened due to heart attacks and strokes.

Behavioral factors such as Unhealthy diet, physical inactivity, tobacco use, use of alcohol, Smoking, Diabetes and physical factors are the main risk factors of heart diseases and strokes. These effects of risk factors may show in increase of blood pressure, blood glucose, and weight gain.

According to WHO among global CVD deaths,80% happens in low and middle income countries. Not identifying the risk factors, having limited access to treatments, not having primary health care programs for early detection might be the reason for this.

By developing evidence based guidelines for prevention CVD, finding the main risk factors of CVD diseases and by having awareness programs of identifying CVD events and how to prevent them can reduce the deaths happens due to CVD events.

According to WHO in Sri Lanka more than 40% of the deaths happened due to CVD in 2021. This synthetically generated data set shows the details of these risk factors before and after the 18 month program initiated by the Ministry of Health Sri Lanka. This program was conducted across three districts Colombo, Kandy and Galle featured health education sessions, lifestyle counseling, and community exercise programs. This study aims to find the effectiveness of the CVD prevention program on major risk factors.

DATA DESCRIPTION

This is a Synthetically generated dataset of 1500 participants and 17 variables. This dataset includes Demographic details such as age, education, gender and district, pre program health details , post program health details, program participation details and whether participant experienced a CVD event during follow-up.

These are the variables included in this dataset.

1. Participant Demographics

- ID: Unique code (CVD001–CVD1500).
- Age: Age of the participant
- Gender: Female or Male
- District: Living area (Colombo, Kandy , Galle)
- Education: Education Status (Primary, Secondary, Higher)

2. Pre-Program Health

- Pre_SBP: Systolic Blood pressure before program (mmHg)
- Pre_BMI: Body mass index value before program (Kg/m²)
- Smoking: Smoking status before program (If smoking -1 , not smoking -0)
- Pre_Exercise: Number of days exercise per week before the program
- Diabetes: Diabetes status (Have diabetes – 1, Don't have diabetes -0)

3. Program Participation

- Participation: By district – If participated to the program or not(1- participated, 0- not participated)
- Sessions: Number of Sessions attended by each participant

4. Post-Program Health

- Post_SBP - Systolic Blood pressure after the program (mmHg)
- Post_BMI - Body mass index value after the program (Kg/m²)
- Post_Smoking – Smoking Status after the program (1=yes, 0- No)
- Post_Exercise – No of days exercise per week after the program
- CVD Event – whether participant experienced a CVD event during follow-up (1-Yes, 0-No)

OBJECTIVES

The primary objective of this study is to find the effectiveness of the CVD prevention program and the health improvements after the prevention program. This study mainly focusses on the below objectives.

- Evaluate the impact of the CVD prevention program on major risk factors
 - Determine if the program helped to reduce systolic blood pressure, BMI, Smoking prevalence and CVD incidence by comparing pre and post program results.
- Analyze whether different districts or age groups experienced varying levels of health improvements.
 - Determine if there is a variation of improvement in risk factors in different age groups, districts, education level and gender.
- Investigate any potential relationships between baseline risk factors (e.g., diabetes, family history) and changes in cardiovascular health outcomes
 - Find the associations between pre-existing risk factors such as diabetes status, smoking behavior, and other baseline characteristics and the health improvements after the program.

This study will present an analysis about the effectiveness of the cardiovascular disease prevention program among 1500 participants across three different districts (Colombo, Kandy, Galle). The findings from this study will help to evaluate the major risk factors and how the program helped to reduce the CVD diseases around the country.

Descriptive Analysis

Descriptive analysis in this report is done using summary statistics, frequency tables, proportion tables, and different types of plots. Below shows the summary of the 8 quantitative and 8 qualitative variables.

```
> summary(dataset)
      ID          Age          Gender          District          Education
Length:1500   Min.   :25.00   Length:1500   Length:1500   Length:1500
Class :character 1st Qu.:42.00   Class :character  Class :character  Class :character
Mode  :character  Mean    :50.56   Mode  :character  Mode  :character  Mode  :character
                  3rd Qu.:60.00
                  Max.    :80.00

      Pre_SBP      Pre_BMI      Smoking      Pre_Exercise      Diabetes      Participation
Min.   :110.0    Min.   :18.50    No :1193    Min.   :0.000    No :1227    No :570
1st Qu.:129.0    1st Qu.:23.70    Yes: 307    1st Qu.:1.000    Yes: 273    Yes:930
Median :141.0    Median :26.60                      Median :2.000
Mean   :142.2    Mean   :26.81                      Mean   :2.017
3rd Qu.:154.0    3rd Qu.:29.70                      3rd Qu.:3.000
Max.   :198.0    Max.   :39.30                      Max.   :4.000

      Sessions      Post_SBP      Post_BMI      Post_Smoking      Post_Exercise
Min.   : 0.000    Min.   :100.0    Min.   :18.50    No :1257    Min.   :0.000
1st Qu.: 0.000    1st Qu.:120.0    1st Qu.:22.40    Yes: 243    1st Qu.:2.000
Median : 7.000    Median :133.0    Median :25.40                      Median :4.000
Mean   : 5.607    Mean   :133.7    Mean   :25.58                      Mean   :3.752
3rd Qu.:10.000    3rd Qu.:146.0    3rd Qu.:28.70                      3rd Qu.:5.000
Max.   :12.000    Max.   :195.0    Max.   :38.70                      Max.   :7.000

      CVD_Event
Length:1500
Class :character
Mode  :character
```

Descriptive statistics for numerical variables

Variable	Mean	Median	Standard Deviation
Age	50.56	50	12.45689
Pre_SBP	142.2	141.0	17.33958
Pre_BMI	26.81	26.60	4.110086
Pre_Exercise	2.017	2.000	1.40631
Sessions	5.607	7.000	4.675596
Post_SBP	133.7	133.0	18.24234
Post_BMI	25.58	25.40	4.157066
Post_Exercise	3.752	4.000	1.895492

According to this data set 8 variables out of 17 were numerical variables and the summary statistics including Mean, Median and standard deviation for those variables is included in the above table. participants ages range from age 25 – 80 years with the Mean of 50.56 years and

median of 50 years. On average each person attended 5.6 sessions out of maximum of 12 sessions in this program.

The Mean Systolic blood pressure before the program is 142.2 mmHg and after the program is 133.7 mmHg which indicates a reduction of systolic blood pressure after the CVD prevention program. SBP values range from 110 – 198 mmHg before the program and 100 – 195 after the program.

Mean BMI of participants before the program is 26.81 Kg^m⁻² and Mean BMI after the program is 25.58 which also shows a reduction after the program. BMI range from 18.5 – 39.3 before the program and 18.5 – 38.7 after the program.

Mean number of days participants were exercising per week before the program is 2.017 and after the program is 3.752 days which shows a increase of days. Number of days exercise before the program ranges from 0 – 4 and after the program range from 0 – 7 which is highlights a good improvement.

Descriptive Statistics for categorical variables

<i>variable</i>	Categories	Frequencies	Proportions
<i>Gender</i>	Female	801	0.534
	Male	699	0.466
<i>District</i>	Colombo	603	0.402
	Galle	369	0.246
	Kandy	528	0.352
<i>Education</i>	Higher	320	0.2133333
	Primary	486	0.3240000
	Secondary	694	0.4626667
<i>Smoking</i>	No	1193	0.7953333
	Yes	307	0.2046667
<i>Diabetes</i>	No	1227	0.818
	Yes	273	0.182
<i>participation</i>	No	570	0.38
	Yes	930	0.62
<i>Post_Smoking</i>	No	1257	0.838
	Yes	243	0.162
<i>CVD_Event</i>	No	1405	0.9366667
	Yes	95	0.06333333

From the variables in this dataset 8 variables out of 17 are categorical and the frequencies of each category and proportions are summarized in the above table.

Among the people in this study more than half (62%) of the people have participated the CVD prevention program and cardiovascular disease occurred only for 95 people out of 1500 which is only 0.06%.

As shown in this table number of females are higher than number of males where 801 are females and 699 are males. In this study 3 districts have been chosen which are Colombo, Galle, and Kandy. Among those three districts most of the people are from Colombo which is 603. From Galle and Kandy there were 369 and 528 people respectively. Education is Categorized in to three groups which are Primary, Secondary and Higher education. Most of the people from this study have up to secondary education which is 46%. Only 21.3% of the people have education up to higher education and 32.4% have up to primary education.

Among the people in this study most of the people (79.5%) are not smoking before the CVD prevention program and after the program people who do not smoke have increased to 83.8%. Only 273 people out of 1500 have diabetes in this study.

Frequency tables

```
> ds_freq_table(dataset, Age, 3)
```

Variable: Age						
Bins			Frequency	Cum Frequency	Percent	Cum Percent
25	-	43.3	461	461	30.73	30.73
43.3	-	61.7	728	1189	48.53	79.27
61.7	-	80	311	1500	20.73	100
Total			1500	-	100.00	-

This table of ages is divided in to three categories which are 25 – 43.3, 43.3 – 61.7 and 61.7 – 80. According to this frequency table most of the people are from 43.3 – 61.7 age group and least number of people are Seniors (61.7 – 80). 48.53% of people are middle aged while only 20.73% are seniors.


```
> ds_freq_table(dataset, Sessions, 4)
Variable: Sessions
```

Bins	Frequency	Cum Frequency	Percent	Cum Percent
0 - 3	570	570	38	38
3 - 6	139	709	9.27	47.27
6 - 9	513	1222	34.2	81.47
9 - 12	537	1759	35.8	117.27
Total	1500	-	100.00	-

Number of sessions were divided in to 4 groups as shown in this frequency table and most of the people(38%) have attended only 0 -3 days. Least number of people have attended 3-6 days which is 9.27%.

In these two frequency tables BMI is divided in to three groups which are 18.5 - 25.4, 25.4 – 32.4 and 32.4 – 39.3. According to WHO 18.5 – 25.5 is Normal weight and over 25.5 is considered as overweight. Therefore the first category of this table can take as the normal weight category and other two categories can be taken as overweight category.

```
> ds_freq_table(dataset, Pre_BMI, 3)
Variable: Pre_BMI
```

Bins	Frequency	Cum Frequency	Percent	Cum Percent
18.5 - 25.4	606	606	40.4	40.4
25.4 - 32.4	745	1351	49.67	90.07
32.4 - 39.3	149	1500	9.93	100
Total	1500	-	100.00	-

The above table is the frequency table for BMI values before the CVD prevention program. According to this table most of the BMI values are over 25.4 and only 40% of the people have normal weight.

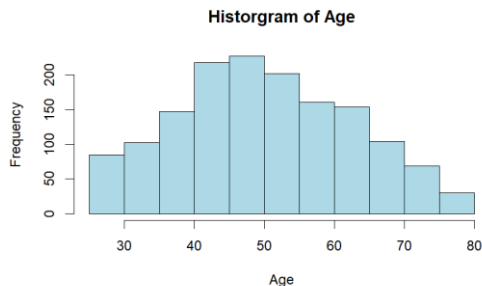
```
> ds_freq_table(dataset, Post_BMI, 3)
```

Variable: Post_BMI						
Bins			Frequency	Cum Frequency	Percent	Cum Percent
18.5	-	25.2	739	739	49.27	49.27
25.2	-	32	655	1394	43.67	92.93
32	-	38.7	106	1500	7.07	100
Total			1500	-	100.00	-

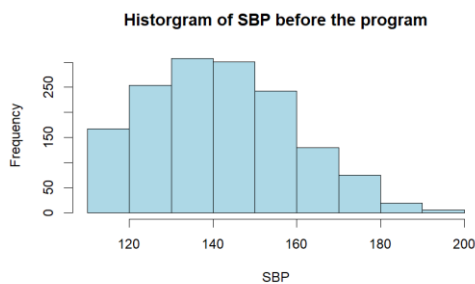
According to this frequency table of post BMI values Normal weight people have increased to 49.27% which shows the progress of the program.

plots

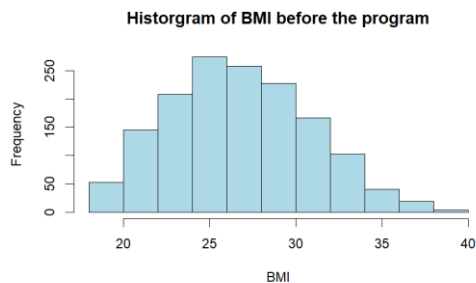
Below are some plots of describing and comparison of the variables.



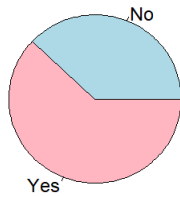
This histogram of age ranges from 25-80 shows a normally distributed plot which indicates that most of the people are near to age 50.



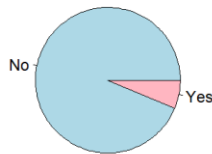
This histogram of Systolic blood pressure shows a right skewed distribution.



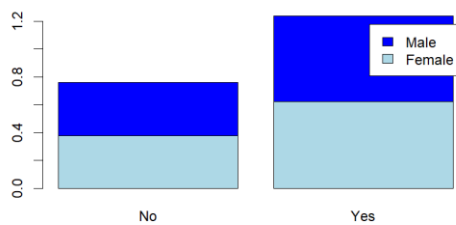
Histogram of BMI before the program shows a right skewed distribution indicating most of the people have lower BMI value

Participation distribution

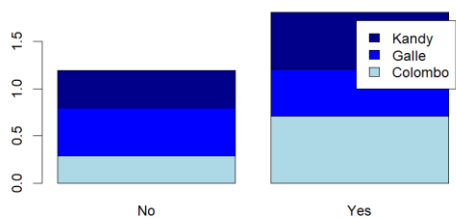
This pie chart shows the distribution of participation (If participated or not). According to this pie chart more than half have been participated to the program.

CVD event distribution

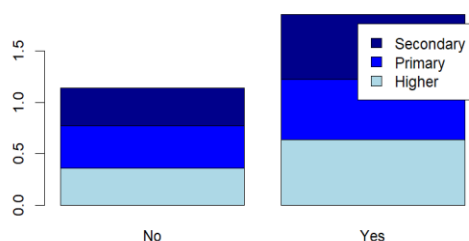
According to this Pie chart of CVD events only small number of people have experienced cardiovascular disease.

Compound Bar chart for participated status and gender

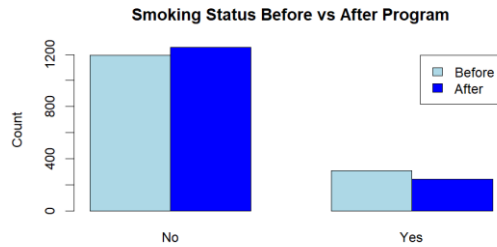
Among the people who participated the CVD prevention program Female count is slightly higher than male count.

Compound Bar chart for participated status and district

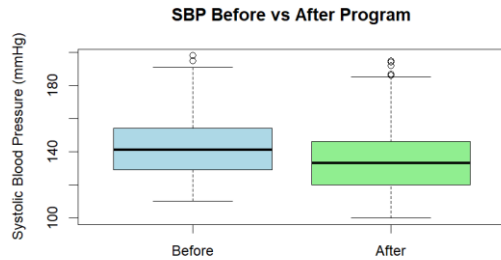
Most of the people are from Colombo among the people who participated the program and least number of people participated the program from Galle.

Compound Bar chart for participated status and education stat

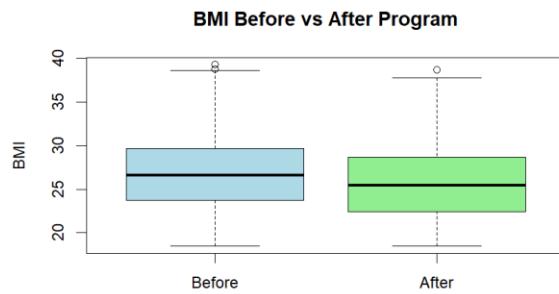
Most of the people among the people who have participated the program have secondary education while least number of people have higher education.



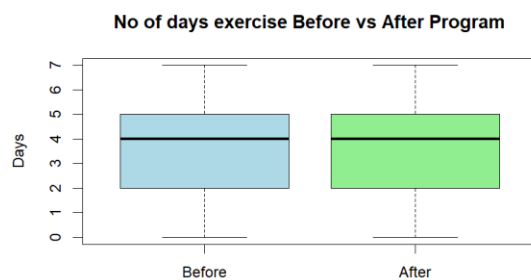
This bar plot compares the smoking status before and after the CVD prevention program. As shown in the plot number of people who are not smoking have been increased after the program.



Median SBP Count have been lower after the prevention program. Also there are some outliers in both before and after Systolic blood pressure.



Median BMI is slightly lower after the prevention program. There are some outliers in BMI in both before and after the program.



Median of number of days after the exercise is slightly higher compared to before.

Inferential Analysis

Checking Normality Assumptions of Variables

To check the normality assumption in order to decide which type of test to use this study (parametric or non parametric tests) Shapiro test was used and the results are shown below.

H0: Data is normally distributed

H1: Data is not normally distributed

```
> shapiro.test(dataset$Age)

      Shapiro-Wilk normality test

data:  dataset$Age
W = 0.98579, p-value = 5.615e-11

> shapiro.test(dataset$Pre_SBP)

      Shapiro-Wilk normality test

data:  dataset$Pre_SBP
W = 0.98567, p-value = 4.902e-11

> shapiro.test(dataset$Pre_BMI)

      Shapiro-Wilk normality test

data:  dataset$Pre_BMI
W = 0.98849, p-value = 1.635e-09

> shapiro.test(dataset$Pre_Exercise)

      Shapiro-Wilk normality test

data:  dataset$Pre_Exercise
W = 0.88962, p-value < 2.2e-16
```

```
> shapiro.test(dataset$Sessions)

      Shapiro-Wilk normality test

data:  dataset$Sessions
W = 0.81703, p-value < 2.2e-16

> shapiro.test(dataset$Post_SBP)

      Shapiro-Wilk normality test

data:  dataset$Post_SBP
W = 0.98736, p-value = 3.784e-10

> shapiro.test(dataset$Post_BMI)

      Shapiro-Wilk normality test

data:  dataset$Post_BMI
W = 0.98175, p-value = 7.322e-13

> shapiro.test(dataset$Post_Exercise)

      Shapiro-Wilk normality test

data:  dataset$Post_Exercise
W = 0.95439, p-value < 2.2e-16
```

Since all the p values obtained from the Shapiro test are less than 0.05 we have to reject null hypothesis. From that we can conclude that data is not normally distributed. Therefore parametric test can not be used in this study and **non-parametric test should be used** to analyze data in this study.

Impact of the CVD prevention program on major risk factors and CVD incidence among community participants

In this section Wilcoxon sign rank test and chi square test is used to find if there is any improvement of the risk factors (SBP, BMI, Exercise days per week) after the program. Below shows the results of those tests.

• Impact of CVD prevention program on SBP

$$H_0: M_{\text{pre_SBP}} - M_{\text{post_SBP}} = 0$$

$M_{\text{Pre_SBP}}$ – Median of SBP value before the program

$$H_1: M_{\text{pre_SBP}} - M_{\text{post_SBP}} > 0$$

$M_{\text{Post_SBP}}$ – Median SBP value after the program

```
> #wilcoxons signed rank test for pre vs post SBP
> wilcox.test(dataset$Pre_SBP, dataset$Post_SBP, paired=TRUE, alternative="greater", conf.level = 0.95)

Wilcoxon signed rank test with continuity correction

data:  dataset$Pre_SBP and dataset$Post_SBP
V = 1000405, p-value < 2.2e-16
alternative hypothesis: true location shift is greater than 0
```

Since p-value < 0.05, you reject H_0 at the 5% significance level.

Interpretation: There is strong evidence that Pre_SBP is significantly greater than Post_SBP.

Conclusion: The program helped in reduction in systolic Blood pressure.

• Impact of CVD prevention program on BMI

$$H_0: M_{\text{pre_BMI}} - M_{\text{post_BMI}} = 0$$

$M_{\text{Pre_BMI}}$ – Median of BMI value before the program

$$H_1: M_{\text{pre_BMI}} - M_{\text{post_BMI}} > 0$$

$M_{\text{Post_BMI}}$ – Median BMI value after the program

```
> #wilcoxons signed rank test for pre vs post Bmi
> wilcox.test(dataset$Pre_BMI, dataset$Post_BMI, paired=TRUE, alternative = "greater", conf.int = 0.95)

Wilcoxon signed rank test with continuity correction

data:  dataset$Pre_BMI and dataset$Post_BMI
V = 1097421, p-value < 2.2e-16
alternative hypothesis: true location shift is greater than 0
95 percent confidence interval:
 1.15001      Inf
sample estimates:
(pseudo)median
 1.200012
```

Since p-value < 0.05, you reject H_0 at the 5% significance level.

Interpretation: There is strong evidence that Pre_BMI is significantly greater than Post_BMI.

Conclusion: The BMI value is reduced after the program.

- **Impact of CVD prevention program on Exercise days per week**

H0: $M_{\text{post_Exercise}} - M_{\text{pre_Exercise}} \leq 0$

$M_{\text{Post_Exercise}}$ – Median Post Exercise days per week

H1: $M_{\text{post_Exercise}} - M_{\text{pre_Exercise}} > 0$

$M_{\text{Pre_Exercise}}$ – Median Pre Exercise days per week

```
> #wilcoxons signed rank test for pre vs post Exercise days
> wilcox.test(dataset$Post_Exercise, dataset$Pre_Exercise, paired=TRUE, alternative="greater", conf.level = 0.95)

Wilcoxon signed rank test with continuity correction

data: dataset$Post_Exercise and dataset$Pre_Exercise
V = 735078, p-value < 2.2e-16
alternative hypothesis: true location shift is greater than 0
```

Since p-value < 0.05, you reject H_0 at the 5% significance level.

Interpretation: There is strong evidence that Post_Exercise is significantly greater than Pre_Exercise.

Conclusion: No of exercise days per week after the program is higher than before the program.

- **Association between CVD event during follow up and Participation Status**

H0: There is no association between CVD_Event and Participation

H1: There is a association between CVD_Event and Participation

```
> # CVD incidence vs participation
> chisq.test(table(dataset$CVD_Event, dataset$Participation), correct = TRUE)

Pearson's Chi-squared test with Yates' continuity correction

data: table(dataset$CVD_Event, dataset$Participation)
X-squared = 0.55141, df = 1, p-value = 0.4577
```

Since p-value < 0.05, you reject H_0 at the 5% significance level.

Interpretation: There is strong evidence to say that there is a association between CVD_Event and Participation

Conclusion: Participation Status and CVD event during the follow up period is related to each other.

Analyze whether different districts or age groups experienced varying levels of health improvements

In this section Kruskal wallis and Wilcoxon test is used to test if the improvements in health measurements (BMI and SPB change) after the CVD prevention program differ in different age groups, districts and gender groups.

- **Health improvement in different districts**

SBP Change

H0: There is no difference in median SBP change among three districts in population

H1: There is a difference in median SBP change in at least one district

```
> SBP_change <- dataset$Pre_SBP - dataset$Post_SBP
> kruskal.test(SBP_change ~ District, data=dataset)

Kruskal-Wallis rank sum test

data: SBP_change by District
Kruskal-Wallis chi-squared = 41.722, df = 2, p-value = 8.715e-10
```

Since $p\text{-value} < 0.05$, you reject H_0 at the 5% significance level.

Conclusion: There is strong evidence to say that there is a difference in median SBP changes in at least one district at 5% significance level.

BMI Change

H0: There is no difference in median BMI change among three districts in population

H1: There is a difference in median BMI change in at least one district

```
> BMI_change <- dataset$Pre_BMI - dataset$Post_BMI
> kruskal.test(BMI_change ~ District, data=dataset)

Kruskal-Wallis rank sum test

data: BMI_change by District
Kruskal-Wallis chi-squared = 15.505, df = 2, p-value = 0.0004296
```


Since $p\text{-value} < 0.05$, you reject H_0 at the 5% significance level.

Conclusion: There is strong evidence to say that there is a difference in median BMI changes in at least one district at 5% significance level.

- **Health improvement in different age groups**

SBP Change

H_0 : There is no difference in median SBP change among the age group in population

H_1 : There is a difference in median SBP change in at least one district

```
> kruskal.test(SBP_change ~ AgeGroup, data = dataset)

Kruskal-wallis rank sum test

data:  SBP_change by AgeGroup
Kruskal-wallis chi-squared = 2.069, df = 2, p-value = 0.3554
```

Since $p\text{-value} > 0.05$, you do not reject H_0 at the 5% significance level.

Conclusion: There is strong evidence to say that there is no difference in median SBP changes in all age groups at 5% significance level.

BMI Change

H_0 : There is no difference in median BMI change among the age groups in population

H_1 : There is a difference in median BMI change in at least one age group

```
> kruskal.test(BMI_change ~ AgeGroup, data = dataset)

Kruskal-wallis rank sum test

data:  BMI_change by AgeGroup
Kruskal-wallis chi-squared = 4.3788, df = 2, p-value = 0.112
```

Since p-value > 0.05, you do not reject H_0 at the 5% significance level.

Conclusion: There is strong evidence to say that there is no difference in median BMI changes in all age groups at 5% significance level.

- **Health improvement in two gender groups**

SBP Change

$H_0: M_{\text{Male}} = M_{\text{Female}}$

M_{Male} – Median SBP change in Males

$H_1: M_{\text{Male}} \neq M_{\text{Female}}$

M_{Female} – Median SBP change in Female

```
> wilcox.test(SBP_change ~ dataset$Gender, data = dataset)

Wilcoxon rank sum test with continuity correction

data: SBP_change by dataset$Gender
W = 283581, p-value = 0.6639
alternative hypothesis: true location shift is not equal to 0
```

Since p-value > 0.05, you do not reject H_0 at the 5% significance level.

Conclusion: There is strong evidence to say that Median SBP changes in both Males and Females are same at 5% significance level.

BMI Change

$H_0: M_{\text{Male}} = M_{\text{Female}}$

M_{Male} – Median BMI change in Males

$H_1: M_{\text{Male}} \neq M_{\text{Female}}$

M_{Female} – Median BMI change in Female

```
> wilcox.test(BMI_change ~ dataset$Gender, data = dataset)

Wilcoxon rank sum test with continuity correction

data: BMI_change by dataset$Gender
W = 283035, p-value = 0.7123
alternative hypothesis: true location shift is not equal to 0
```

Since $p\text{-value} > 0.05$, you do not reject H_0 at the 5% significance level.

Conclusion: There is strong evidence to say that Median BMI changes in both Males and Females are same at 5% significance level.

Investigate any potential relationships between baseline risk factors (e.g., diabetes, family history) and program effectiveness

In this part of Wilcoxon test and chi square test is used to find if there is any relationship between baseline risk factors and improvements of health factors due to the program.

- **Effect of Diabetes on Systolic Blood Pressure**

H_0 : There is no difference in SBP change between people with and without diabetes.

H_1 : There is a difference in SBP change between people with and without diabetes.

```
> wilcox.test(SBP_change ~ Diabetes, data=dataset)
```

```
Wilcoxon rank sum test with continuity correction
```

```
data: SBP_change by Diabetes
```

```
W = 168156, p-value = 0.9174
```

```
alternative hypothesis: true location shift is not equal to 0
```

Since $p\text{-value} > 0.05$, you do not reject H_0 at the 5% significance level.

Conclusion: There is strong evidence to say that there is no difference in SBP change between people with and without diabetes at 5% significant level.

- **Effect of Diabetes on BMI**

H_0 : There is no difference in BMI change between people with and without diabetes.

H_1 : There is a difference in BMI change between people with and without diabetes.

```
> wilcox.test(BMI_change ~ Diabetes, data=dataset)

Wilcoxon rank sum test with continuity correction

data: BMI_change by Diabetes
W = 168508, p-value = 0.8745
alternative hypothesis: true location shift is not equal to 0
```

Since p-value > 0.05, you do not reject H_0 at the 5% significance level.

Conclusion: There is strong evidence to say that there is no difference in BMI change between people with and without diabetes at 5% significant level.

- **Effect of Diabetes on Cardiovascular Event during follow up**

H_0 : CVD_Event and Diabetes are independent

H_1 : CVD_Event and Diabetes are not independent

```
> chisq.test(table(dataset$CVD_Event, dataset$Diabetes))

Pearson's Chi-squared test with Yates' continuity correction

data: table(dataset$CVD_Event, dataset$Diabetes)
X-squared = 6.403, df = 1, p-value = 0.01139
```

Since p-value < 0.05, you reject H_0 at the 5% significance level.

Conclusion: There is strong evidence to say that CVD event during follow up and having diabetes are not independent.

- **CVD event vs smoking**

H_0 : CVD_Event and Smoking are independent

H_1 : CVD_Event and smoking are not independent

```
> chisq.test(table(dataset$CVD_Event, dataset$Smoking))
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(dataset$CVD_Event, dataset$Smoking)
X-squared = 20.086, df = 1, p-value = 7.405e-06
```

Since $p\text{-value} < 0.05$, you reject H_0 at the 5% significance level.

Conclusion: There is strong evidence to say that CVD event during follow up and smoking status are not independent.

- **SBP Change vs Education level**

H_0 : There is no difference in median SBP change among the Education levels

H_1 : There is a difference in median SBP change in at least one Education level

```
> kruskal.test(SBP_change ~ Education, data = dataset)
```

Kruskal-Wallis rank sum test

```
data: SBP_change by Education
Kruskal-Wallis chi-squared = 4.1347, df = 2, p-value = 0.1265
```

Since $p\text{-value} > 0.05$, you do not reject H_0 at the 5% significance level.

Conclusion: There is strong evidence to say that there is no difference in median SBP changes in all Education levels at 5% significance level.

- **BMI Change vs Education level**

H_0 : There is no difference in median BMI change among the Education levels

H_1 : There is a difference in median BMI change in at least one Education level

```
> kruskal.test(BMI_change ~ Education, data = dataset)
```

```
Kruskal-wallis rank sum test
```

```
data: BMI_change by Education
```

```
Kruskal-wallis chi-squared = 1.0441, df = 2, p-value = 0.5933
```

Since p-value > 0.05, you do not reject H_0 at the 5% significance level.

Conclusion: There is strong evidence to say that there is no difference in median BMI changes in all Education levels at 5% significance level.

Conclusion

This study was done to investigate the impact of the 18 months community based cardiovascular disease (CVD) prevention program across three districts (Colombo, Kandy, and Galle) initiated by Ministry of Health, Sri Lanka. This report analysis cardiovascular risk factors, lifestyle behaviors, and demographic characteristics of the community participants.

The program shows a reduction of systolic blood pressure, BMI value, Smoking prevalence and increase in exercise days per week. All these improvements of risk factors and lifestyle behaviors show the effectiveness of CVD prevention program.

When comparing outcomes across districts both BMI and SBP reduction is significantly different in at least one district. This might be due to participation status, or access to health care facilities. Reduction of BMI and SBP values are same in all age groups and gender. This shows that this program can be applicable for any age group or gender.

It was found that there is a relationship between CVD event occurring during follow up and diabetes proving the clinical evidence. Also, there is no relationship between diabetes and BMI, SBP improvements indicating that the program effectiveness is same for diabetic and non diabetic individuals.

Overall, this program has a positive effect of cardiovascular disease prevention. Future analysis by finding new risk factors that affect the cardiovascular health might help in reduction in cardiovascular diseases and improve the health of the community.

Dataset - https://drive.google.com/file/d/1x1OXHa_l8hA3ZXHpR5lCKdXQgAGdtTzt/view?usp=sharing

R Codes

R codes used to generate the Dataset

```

library(dplyr)
library(truncnorm)
set.seed(16807)

n <- 1500

ID <- sprintf("CVD%03d", 1:n)

Age <- round(rtruncnorm(n, a = 25, b = 80, mean = 50, sd = 15))

Gender <- sample(c("Female", "Male"), n, replace = TRUE, prob = c(0.52, 0.48))

District <- sample(c("Colombo", "Kandy", "Galle"), n, replace = TRUE, prob = c(0.40, 0.35, 0.25))

Education <- sapply(District, function(d) {
  if (d == "Colombo") sample(c("Primary", "Secondary", "Higher"), 1, prob = c(0.20, 0.50, 0.30))
  else if (d == "Kandy") sample(c("Primary", "Secondary", "Higher"), 1, prob = c(0.35, 0.45, 0.20))
  else sample(c("Primary", "Secondary", "Higher"), 1, prob = c(0.45, 0.40, 0.15))
})

Pre_SBP <- round(rtruncnorm(n, a = 110, b = 200, mean = 140, sd = 20))

Pre_BMI <- round(rtruncnorm(n, a = 18.5, b = 40, mean = 26.5, sd = 4.5), 1)

Smoking <- ifelse(Gender == "Male",
  rbinom(n, 1, 0.35),
  rbinom(n, 1, 0.08))

Pre_Exercise <- sample(0:4, n, replace = TRUE)

Diabetes <- ifelse(Age > 50, rbinom(n, 1, 0.25), rbinom(n, 1, 0.12))

Participation <- mapply(function(d) {

```

```

if (d == "Colombo") rbinom(1,1,0.70)
else if (d == "Kandy") rbinom(1,1,0.60)
else rbinom(1,1,0.50)
}, District)

```

```
Sessions <- ifelse(Participation==1, sample(6:12, n, replace = TRUE), 0)
```

```

BP_improvement <- ifelse(Participation==1, sample(5:20, n, replace=TRUE), sample(0:5, n,
replace=TRUE))
Post_SBP <- pmax(100, Pre_SBP - BP_improvement)

```

```

BMI_improvement <- ifelse(Participation==1, runif(n,0.5,3.0), runif(n,0,1.0))
Post_BMI <- round(pmax(18.5, Pre_BMI - BMI_improvement),1)

```

```

Post_Smoking <- sapply(1:n, function(i){
  if (Smoking[i]==1 & Participation[i]==1) {
    if (rbinom(1,1,0.30)==1) return(0) else return(1)
  } else if (Smoking[i]==1 & Participation[i]==0) {
    if (rbinom(1,1,0.10)==1) return(0) else return(1)
  } else {
    return(0)
  }
})

```

```

Exercise_increase <- ifelse(Participation==1, sample(1:4, n, replace=TRUE), sample(0:1, n,
replace=TRUE))
Post_Exercise <- pmin(7, Pre_Exercise + Exercise_increase)

```

```
CVD_prob <- rep(0.03, n)
```

```

CVD_prob <- CVD_prob + ifelse(Age>60, 0.04, 0)
CVD_prob <- CVD_prob + ifelse(Gender=="Male", 0.03, 0)
CVD_prob <- CVD_prob + ifelse(Post_Smoking==1, 0.05, 0)
CVD_prob <- CVD_prob + ifelse(Diabetes==1, 0.06, 0)
CVD_prob <- CVD_prob + ifelse(Post_SBP>160, 0.04, 0)
CVD_prob <- CVD_prob - ifelse(Participation==1, 0.02, 0)

```

```
CVD_prob <- pmin(pmax(CVD_prob,0),1)
```



```

CVD_Event <- ifelse(runif(n) < CVD_prob, "Yes", "No")

df <- data.frame(
  ID, Age, Gender, District, Education,
  Pre_SBP, Pre_BMI, Smoking, Pre_Exercise, Diabetes,
  Participation, Sessions,
  Post_SBP, Post_BMI, Post_Smoking, Post_Exercise,
  CVD_Event
)

head(df)

# Export to CSV
write.csv(df, "C:/Users/oshin/Desktop/health assignment/CVD_dataset.csv", row.names = FALSE)

```

R codes used to analyze the Dataset

```

dataset= read.csv("C:\\Users\\oshin\\Desktop\\health assignment\\CVD_dataset.csv")
dataset$Smoking <- factor(dataset$Smoking, labels = c("No", "Yes"))
dataset$Diabetes <- factor(dataset$Diabetes, labels = c("No", "Yes"))
dataset$Participation <- factor(dataset$Participation, labels = c("No", "Yes"))
dataset$Post_Smoking <- factor(dataset$Post_Smoking, labels = c("No", "Yes"))

summary(dataset)

#Standard deviations of numerical variables
sd(dataset$Age)
sd(dataset$Pre_SBP)
sd(dataset$Pre_BMI)
sd(dataset$Pre_Exercise)
sd(dataset$Sessions)
sd(dataset$Post_SBP)
sd(dataset$Post_BMI)
sd(dataset$Post_Exercise)

```

```
#mean of numerical variables
mean(dataset$Age)
mean(dataset$Pre_SBP)
mean(dataset$Pre_BMI)
mean(dataset$Pre_Exercise)
mean(dataset$Sessions)
mean(dataset$Post_SBP)
mean(dataset$Post_BMI)
mean(dataset$Post_Exercise)

#proportion tables and frequency tables for categorical variables
table(dataset$Gender)
prop.table(table(dataset$Gender))

table(dataset$District)
prop.table(table(dataset$District))

table(dataset$Education)
prop.table(table(dataset$Education))

table(dataset$Smoking)
prop.table(table(dataset$Smoking))

table(dataset$Diabetes)
prop.table(table(dataset$Diabetes))

table(dataset$Post_Smoking)
prop.table(table(dataset$Post_Smoking))

table(dataset$CVD_Event)
prop.table(table(dataset$CVD_Event))

table(dataset$Participation)
prop.table(table(dataset$Participation))

#frequency tables

library(descriptr)
ds_freq_table(dataset, Age, 3)
ds_freq_table(dataset, Pre_BMI, 3)
```

```

ds_freq_table(dataset, Post_BMI, 3)
ds_freq_table(dataset, Sessions, 4)

##discriptive data analysis
hist(dataset$Age,xlab="Age",ylab="Frequency",col="lightblue",main="Histogram of Age")
hist(dataset$Pre_SBP,xlab="SBP",ylab="Frequency",main="Histogram of SBP before the
program", col ="lightblue")
hist(dataset$Pre_BMI,xlab="BMI",ylab="Frequency",main="Histogram of BMI before the
program", col ="lightblue")
hist(dataset$Pre_Exercise,breaks = seq(-0.5, 4.5, 1),col = "skyblue",main = "Histogram of number
of days exercised before the program",xlab = "No of days",ylab = "Frequency")

pie(table(dataset$Participation),main = "Participation distribution",col =
c("lightblue","lightpink"))
pie(table(dataset$CVD_Event),main = "CVD event distribution",col = c("lightblue","lightpink"))

table_participated=table(dataset$Gender,dataset$Participation)
table_participated=prop.table(table_participated,1)
table_participated
barplot(table_participated,legend.text=row.names(table_participated),main='Compound Bar
chart for participated status and gender',col=c('lightblue','blue'))

table_participated_district=table(dataset$District,dataset$Participation)
table_participated_district=prop.table(table_participated_district,1)
table_participated_district
barplot(table_participated_district,legend.text=row.names(table_participated_district),main='Co
mpound Bar chart for participated status and district',col=c('lightblue','blue','darkblue'))

table_participated_education=table(dataset$Education,dataset$Participation)
table_participated_education
table_participated_education=prop.table(table_participated_education,1)
table_participated_education
barplot(table_participated_education,legend.text=row.names(table_participated_education),ma
in='Compound Bar chart for participated status and education
status',col=c('lightblue','blue','darkblue'))

table_participated_CVD=table(dataset$CVD_Event,dataset$Participation)
table_participated_CVD
table_participated_CVD=prop.table(table_participated_CVD,1)
table_participated_CVD

```

```
barplot(table_participated_CVD,legend.text=c("CVD-yes","CVD-no"),main='Compound Bar chart
for participated status and CVD status',col=c('lightblue','blue','darkblue'))
```

```
smoke_table <- rbind(table(dataset$Smoking),table(dataset$Post_Smoking))
smoke_table
barplot(smoke_table, beside=TRUE,col=c("lightblue","blue"),main="Smoking Status Before vs
After Program",ylab="Count",legend.text = c("Before","After"))
```

```
boxplot(dataset$Pre_SBP, dataset$Post_SBP,names=c("Before",
"After"),col=c("lightblue","lightgreen"),main="SBP Before vs After Program",ylab="Systolic Blood
Pressure (mmHg)")
boxplot(dataset$Pre_BMI, dataset$Post_BMI,names=c("Before",
"After"),col=c("lightblue","lightgreen"),main="BMI Before vs After Program",ylab="BMI")
boxplot(dataset$Post_Exercise, dataset$Post_Exercise,names=c("Before",
"After"),col=c("lightblue","lightgreen"),main="No of days exercise Before vs After
Program",ylab="Days")
```

```
table(dataset$Gender,dataset$Participation)
```

```
#Shapiro test to normality assumption
```

```
shapiro.test(dataset$Age)
shapiro.test(dataset$Pre_SBP)
shapiro.test(dataset$Pre_BMI)
shapiro.test(dataset$Pre_Exercise)
shapiro.test(dataset$Sessions)
shapiro.test(dataset$Post_SBP)
shapiro.test(dataset$Post_BMI)
shapiro.test(dataset$Post_Exercise)
```

```
#wilcoxons signed rank test for pre vs post SBP
```

```
wilcox.test(dataset$Pre_SBP, dataset$Post_SBP, paired=TRUE, alternative="greater",conf.level =
0.95)
```

```
#wilcoxons signed rank test for pre vs post Bmi
```

```
wilcox.test(dataset$Pre_BMI, dataset$Post_BMI, paired=TRUE, alternative = "greater",conf.int =
0.95)
```

```
#wilcoxons signed rank test for pre vs post Exercise days
```

```
wilcox.test(dataset$Post_Exercise, dataset$Pre_Exercise, paired=TRUE,
alternative="greater",conf.level = 0.95)
```

```
# CVD incidence vs participation
chisq.test(table(dataset$CVD_Event, dataset$Participation),correct = TRUE)

#health improvement in different districts
SBP_change <- dataset$Pre_SBP - dataset$Post_SBP
kruskal.test(SBP_change ~ District, data=dataset)

BMI_change <- dataset$Pre_BMI - dataset$Post_BMI
kruskal.test(BMI_change ~ District, data=dataset)

#health improvement in different age groups
dataset$AgeGroup <- cut(dataset$Age, breaks = c(0, 39, 59, 100),labels = c("Young (<40)",
"Middle (40-59)", "Older (60+)"))
table(dataset$AgeGroup)

kruskal.test(SBP_change ~ AgeGroup, data = dataset)
kruskal.test(BMI_change ~ AgeGroup, data = dataset)

#health improvement in gender
wilcox.test(SBP_change ~ dataset$Gender, data = dataset)
wilcox.test(BMI_change ~ dataset$Gender, data = dataset)

#program effect on diabetics (baseline factor)
wilcox.test(SBP_change ~ Diabetes, data=dataset)
wilcox.test(BMI_change ~ Diabetes, data=dataset)

#CDV event and diabatisie
chisq.test(table(dataset$CVD_Event, dataset$Diabetes))

# Smoking vs CVD Event
chisq.test(table(dataset$CVD_Event, dataset$Smoking))

#Education level vs health change
kruskal.test(SBP_change ~ Education, data = dataset)
kruskal.test(BMI_change ~ Education, data = dataset)
```